

NETWORK DESIGN AND IMPLEMENTATION

Multi-Site Healthcare Network

Cisco Packet Tracer Case Study

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Computer Networks / Networking Case Study
Academic Project Documentation

Table of Contents

1. Introduction
2. Project Objectives
3. Network Requirements
4. Network Topology
5. Device Inventory
6. IP Addressing Scheme
7. VLAN Design
8. Router Configuration
9. Inter-Router Connections
10. Static Routing
11. End Device Configuration
12. Network Testing
13. Troubleshooting
14. Final Network Status
15. Conclusion

1. Introduction

This project presents the design and implementation of a multi-site healthcare network using Cisco Packet Tracer. The network connects three locations: Headquarters (HQ), North, and South. The HQ site contains separate departments for Patient Records, Orthodontic Imaging, IT and Billing, and Front Desk.

VLANs are used to logically separate the HQ departments. Router-on-a-Stick provides inter-VLAN communication, while serial WAN links connect the three routers. Static routing is configured to allow communication between all sites.

2. Project Objectives

- Design a multi-site healthcare network.
- Connect HQ, North, and South locations.
- Separate HQ departments using VLANs.
- Configure 802.1Q trunking.
- Implement Router-on-a-Stick inter-VLAN routing.
- Configure serial WAN connections.
- Implement static routing.
- Configure IP addresses and default gateways.

- Test end-to-end connectivity using ping.
- Troubleshoot routing and connectivity problems.

3. Network Requirements

- Three routers: HQ-R1, NORTH-R1, SOUTH-R1.
- Three switches: HQ-SW1, NORTH-SW1, SOUTH-SW1.
- Eight representative PCs at HQ.
- Two PCs at North.
- Two PCs at South.
- VLAN-based department separation at HQ.
- Serial leased-line connections between sites.
- IPv4 addressing using VLSM-style subnet allocation.
- Static routing between the sites.

4. Network Topology

The topology consists of three sites. HQ is the central location. HQ-R1 connects to NORTH-R1 and SOUTH-R1 using serial links. Each router connects to its local switch. The HQ switch carries four department VLANs over a trunk connection to HQ-R1.

The topology screenshot should be inserted here from Cisco Packet Tracer.

5. Device Inventory

Device	Name	Location	Purpose
Router	HQ-R1	HQ	Inter-VLAN routing and WAN connectivity
Router	NORTH-R1	North	North LAN and WAN connectivity
Router	SOUTH-R1	South	South LAN and WAN connectivity
Switch	HQ-SW1	HQ	Connects HQ VLANs and end devices
Switch	NORTH-SW1	North	Connects North PCs
Switch	SOUTH-SW1	South	Connects South PCs

6. IP Addressing Scheme

Network	Subnet	Gateway	Mask	VLAN
Patient Records	192.168.60.0/26	192.168.60.1	255.255.255.192	10
Orthodontic Imaging	192.168.60.64/28	192.168.60.65	255.255.255.240	20
IT & Billing	192.168.60.80/28	192.168.60.81	255.255.255.240	30
Front Desk	192.168.60.96/29	192.168.60.97	255.255.255.248	40
North LAN	192.168.60.104/29	192.168.60.105	255.255.255.248	50
South LAN	192.168.60.112/29	192.168.60.113	255.255.255.248	60
HQ-North WAN	192.168.60.120/3	HQ .121 /	255.255.255.252	WAN

	0	North .122		
HQ-South WAN	192.168.60.124/30	HQ .125 / South .126	255.255.255.252	WAN

7. VLAN Design

VLAN ID	VLAN Name	Department/Site
10	PATIENT_RECORDS	HQ Patient Records
20	ORTHODONTIC_IMAGING	HQ Orthodontic Imaging
30	IT_BILLING	HQ IT & Billing
40	FRONT_DESK	HQ Front Desk
50	NORTH_LAN	North Site
60	SOUTH_LAN	South Site

HQ-SW1 uses Gi0/1 as an 802.1Q trunk toward HQ-R1. North and South use single LAN VLANs because each branch has one local network.

8. Router Configuration

HQ-R1 uses Router-on-a-Stick. Subinterfaces G0/0.10, G0/0.20, G0/0.30, and G0/0.40 provide the default gateways for the four HQ VLANs.

HQ-R1 also uses S0/0/0 for the North WAN link and S0/0/1 for the South WAN link. The HQ serial interfaces were configured as DCE where required and a clock rate was applied.

NORTH-R1 and SOUTH-R1 provide the default gateways for their respective LANs and use serial interfaces for WAN connectivity.

9. Inter-Router Connections

- HQ-R1 S0/0/0 ↔ NORTH-R1 S0/0/0
- HQ-R1 S0/0/1 ↔ SOUTH-R1 S0/0/0
- HQ-North WAN addresses: 192.168.60.121/30 and 192.168.60.122/30
- HQ-South WAN addresses: 192.168.60.125/30 and 192.168.60.126/30

10. Static Routing

Static routes were configured on all three routers so that remote LANs could be reached through the appropriate next-hop router. This provides communication between HQ, North, and South without using a dynamic routing protocol.

11. End Device Configuration

Device	IP Address	Mask	Gateway
PR-PC1	192.168.60.2	/26	192.168.60.1
PR-PC2	192.168.60.3	/26	192.168.60.1
IMG-PC1	192.168.60.66	/28	192.168.60.65
IMG-PC2	192.168.60.67	/28	192.168.60.65
IT-PC1	192.168.60.82	/28	192.168.60.81
IT-PC2	192.168.60.83	/28	192.168.60.81

FD-PC1	192.168.60.98	/29	192.168.60.97
FD-PC2	192.168.60.99	/29	192.168.60.97
NORTH-PC1	192.168.60.106	/29	192.168.60.105
NORTH-PC2	192.168.60.107	/29	192.168.60.105
SOUTH-PC1	192.168.60.114	/29	192.168.60.113
SOUTH-PC2	192.168.60.115	/29	192.168.60.113

12. Network Testing

Connectivity was verified using ping tests from PCs and routers. Tests included local gateway connectivity, inter-site connectivity, and communication between HQ, North, and South.

- South PC to South gateway: successful.
- South PC to North router: successful after static route correction.
- South PC to North PCs: successful.
- South PC to HQ imaging PC: successful.
- HQ imaging PC to North PC: successful.
- Final end-to-end connectivity was verified with 0% packet loss in successful tests.

Packet Tracer ping screenshots should be inserted in this section.

13. Troubleshooting

- A routing issue initially prevented South from reaching the North router. A static route to the HQ-North serial network was added on SOUTH-R1.
- SOUTH-R1 initially had an overlapping serial IP address on S0/0/1. The unused interface was cleared and shut down.
- Initial first-packet ping failures were observed during ARP/path learning. Repeated tests completed successfully.
- Switch VLAN assignments and trunking were verified using show vlan brief and show interfaces trunk.

14. Final Network Status

- All required routers and switches are connected.
- HQ department VLANs are configured.
- HQ trunking is operational.
- Router subinterfaces are up/up.
- WAN serial interfaces are operational.
- Static routing is configured.
- End-to-end connectivity has been tested.
- Configuration was saved using copy running-config startup-config.

15. Conclusion

The multi-site healthcare network was successfully designed and implemented in Cisco Packet Tracer. VLANs provide logical separation between HQ departments, while Router-on-a-Stick enables inter-VLAN communication. Serial WAN links connect HQ with the North and South

sites, and static routing enables communication across the complete network. Connectivity tests confirmed that the final network configuration supports communication between the different departments and locations.